TripMate AI - Project Overview

A multi-agent, Responsible-AI travel planner for worldwide trips

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1. Executive summary

TripMate AI takes a traveller's request in plain English - for example *"Plan 5 days in Jaipur from Delhi, heritage and food, mid-range budget"* - and produces a complete, practical travel plan: flights or trains, hotels, weather and packing advice, a day-by-day itinerary, and an estimated budget.

Instead of one large AI doing everything, the work is split across **a team of specialised AI agents**, each an expert in one area, coordinated automatically. Safety and **Responsible AI** controls are applied at every step - not as an afterthought.

The entire system runs on **free, open tools and models** (a local Llama model and free public APIs), so there is **no per-query cost** and user data can stay on-premise.

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2. How many agents, and what each one does

The system uses **7 AI agents** plus **2 control gates**.

The 7 AI agents

#	Agent	Purpose	Data / tools it uses
1	**Supervisor Agent**	The "brain". Understands the request, extracts the trip details (cities, dates, budget, travellers), and **decides which specialist agents are actually needed** for this trip. No fixed script.	The LLM
2	**Flight Agent**	Finds the best flights and trains, compares cost/time, gives booking tips.	AviationStack (schedules) + web search, via MCP
3	**Hotel Agent**	Recommends and compares hotels across budget levels and suggests the best area to stay.	Web search + Google Places, via MCP
4	**Weather Agent**	Gives a weather outlook and a packing list tailored to the destination and season.	Open-Meteo (free weather API), via MCP
5	**Budget Agent**	Produces an itemised cost estimate in INR and money-saving tips.	The LLM + the other agents' findings

#	Agent	Purpose	Data / tools it uses
6	**Itinerary Agent**	Builds a realistic day-by-day plan of places and activities, grouped to reduce travel.	Web search + Google Places + the weather outlook
7	**Final Response Agent**	Combines all the specialist reports into one clean, readable travel plan.	The LLM

The 2 control gates (not LLM "experts", but essential steps)

Gate	Purpose
Input Guardrail	The security checkpoint at the entrance - validates every request before any work starts.
Human-in-the-Loop	Optional pause for a person to review and approve the plan before the agents run.

Why a team instead of one AI? Each agent has a focused job and the right tool, which makes results more accurate and easier to trust, debug and improve. The Supervisor only runs the agents a given trip needs (e.g. a weekend trip with no origin city skips the Flight Agent), which saves time and compute.

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3. The workflow, step by step - and where guardrails & Responsible AI apply

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STEP 1   User request
            |
            v
STEP 2   INPUT GUARDRAIL          <-- Guardrail + Responsible AI (safety, injection, scope)
            |  (pass)                   If blocked: stop with a clear reason.
            v
STEP 3   SUPERVISOR AGENT          <-- Scope check (worldwide; optional India-only)
            |                          Decides which specialists to run.
            v
STEP 4   HUMAN REVIEW (optional) <-- Human oversight: approve / cancel / adjust
            |  (approved)
            v
STEP 5   SPECIALIST AGENTS         <-- Responsible AI guard on EVERY agent's output
         (Flight, Hotel, Weather,      (PII redaction, safety scan, disclaimers)
          Budget, Itinerary - run
          in parallel)
            |
            v
STEP 6   FINAL RESPONSE AGENT      <-- Responsible AI (final safety pass + PII redaction)
            |
            v
STEP 7   Plan shown to the user  +  a Responsible-AI report of every check that ran
```

Key point for review: a guardrail runs at the entrance (Step 2), the scope policy is enforced at the Supervisor (Step 3), a human can approve (Step 4), and a Responsible-AI guard wraps **every single agent** (Steps 5-6). Safety is layered through the whole pipeline, not bolted on at one point.

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4. Guardrails - where and what

Guardrails are applied in **two layers**, so nothing unsafe gets in and nothing unsafe gets out.

Layer 1 - Input Guardrail (Step 2, at the entrance)

Checks every request before any processing: - **Relevance** - is this actually a travel request? Irrelevant messages are declined. - **Safety** - blocks unsafe, harmful or illegal requests. - **Prompt-injection / jailbreak defense** - detects attempts like *"ignore your instructions"* and blocks them. - **Validity** - empty or nonsensical input is rejected with guidance.

If a request fails, the system **stops and explains why**, rather than guessing.

Layer 2 - Policy Guardrail (Step 3, at the Supervisor)

- **Scope policy: worldwide (configurable).** Trips to any country are supported; an optional India-only mode politely declined and domestic alternatives are suggested.

Output guardrail on every agent (Steps 5-6)

Every specialist agent and the final response pass through an output guardrail (see Responsible AI below).

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5. Responsible AI - the four pillars and where each is used

Responsible AI is built into the system, applied at the steps shown.

Pillar	What it does	Where it runs
Safety / content moderation	Screens content against safety categories using **Llama Guard** (a free, industry-standard safety model from Meta), with a fast rule-based scan as backup.	Input (Step 2) and final output (Step 6); rule scan on every agent (Step 5)
Privacy / PII protection	Automatically detects and redacts personal data - email, phone, Aadhaar, PAN, card numbers - so it is never shown or stored. Prices and dates are kept.	Every agent's output (Steps 5-6) and the stored chat history
Prompt-injection defense	Detects and blocks jailbreak / instruction-override attempts.	Input (Step 2)
Transparency & fairness	A visible "Responsible AI" report shows every check that ran; agents are instructed to be inclusive and non-discriminatory; plans carry a "figures are estimates" disclaimer so users aren't misled.	Every agent (prompt-level) + a transparency panel in the UI

Efficient by design: the per-agent privacy/safety checks are instant rule-based scans (no extra AI calls). The heavier Llama Guard model runs only at the two ends (entrance and final answer), so safety adds at most ~2 extra model calls to an entire plan.

Every check is recorded. The user (and we) can see, per agent, which engine ran, whether any PII was redacted, whether injection was detected, and whether a disclaimer was added - full transparency.

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6. Design highlights for management

- **Zero running cost.** Uses a free local LLM (Ollama / Llama 3) or the free Groq tier, and free public APIs (Open-Meteo for weather). No paid API is required to operate.
- **Privacy-friendly.** Can run entirely on local infrastructure; PII is redacted automatically; user accounts use salted, hashed passwords (never stored in plain text).
- **Resilient.** If an optional data source is unavailable, the affected agent degrades gracefully and the plan is still produced - no hard failures.
- **Memory.** Remembers each user's preferences (e.g. "vegetarian", "prefers trains") across conversations.
- **Human oversight.** A human can review and approve the plan before it is generated (configurable).
- **Automated evaluation.** Every plan is scored (0-100 + grade) by heuristic metrics plus an optional LLM-as-judge, shown in the UI.
- **Modern architecture.** Built on **Google ADK** (agent orchestration, sessions, memory) with tools exposed through **MCP** (Model Context Protocol), the emerging open standard for connecting AI to tools.

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7. Technology at a glance

Area	Choice
Agent orchestration	Google ADK
AI model (LLM)	Llama 3 - local via Ollama (free), or Groq free tier
Safety model	Llama Guard (free)
Tool protocol	MCP (Model Context Protocol)
Weather	Open-Meteo (free, no key)
Flights / search / places	AviationStack, Tavily, Google Places (optional)
Memory	PostgreSQL (or in-memory)
User interface	Streamlit web app (with login)
Accounts	SQLite + PBKDF2 password hashing

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8. Summary - the one-line answers

- **How many agents?** 7 AI agents (1 Supervisor + 5 specialists + 1 Final Response), plus 2 control gates (Input Guardrail and Human Review).
- **What is each agent for?** See Section 2 - flights, hotels, weather, budget, itinerary, with a Supervisor that routes and a Final agent that assembles the answer.
- **Where are guardrails used?** At the entrance (safety, injection, relevance), at the Supervisor (scope policy), and on every agent's output.
- **Where is Responsible AI used?** Safety moderation at the entrance and final output, PII redaction on every agent, injection defense at the entrance, and transparency/fairness throughout - with a visible report of every check.

TripMate AI - a safe, explainable, cost-free multi-agent travel planner.